

The C*-algebra generated by the CAR representation and the shell unitaries

`MathlibAnnex.CStarAlgebra.CAR.AtomicCounterexampleAlgebra`

abbrev

Defines the single algebra T used in the irreducible and separable faithful representations.

Statement

Let A be the completed CAR algebra and let ω be its distinguished pure state. Let I be the set of unitary-equivalence classes of pure-state GNS representations. Choose a representative $\pi_i : A \rightarrow B(H_i)$ in each class, keeping the GNS representation of ω in its distinguished class, and put $\mathcal{H} = \bigoplus_{i \in I} H_i$ and $\pi = \bigoplus_{i \in I} \pi_i$. Fix the automorphisms and shell partial isometries supplied by CAR homogeneity, and choose the corresponding unitaries L_i on $\mathcal{H}$ by the cited shell construction. Define

$$T = C^*(\pi(A) \cup \{L_i : i \in I\}) \subseteq B(\mathcal{H}).$$

Definition

T is the norm-closed unital complex *-subalgebra of $B(\mathcal{H})$ generated by $\pi(A)$ and the unitaries L_i . The automorphisms, shell partial isometries and resulting unitaries are chosen once; the abbreviation does not choose another isomorphic copy of the algebra.

Assumptions

The CAR algebra, the pure state ω , the GNS representatives and the unitaries L_i are those in the cited fixed construction. The index set I is not assumed countable.

Conclusion

The notation T refers to this particular generated C*-algebra. It does not denote a new isomorphic copy when different representations of T are considered.

The abbreviation defines the algebra; separate theorems establish its simplicity, uniqueness of the nonzero irreducible representation up to unitary equivalence, and failure to be isomorphic to an algebra of compact operators. The fixed construction uses the proved CAR homogeneity theorem, without an additional KOS or CH hypothesis.

Main citations

- Definition and its exact construction —
`MathlibAnnex.CStarAlgebra.CAR.AtomicCounterexampleAlgebra`

- The once-chosen representative shell family — `MathlibAnnex.CStarAlgebra.CAR.homogeneityShellFamily`
- The generated algebra for a fixed shell family — `MathlibAnnex.CStarAlgebra.CAR.ShellFamilyTarget`
- The fixed ambient link operators — `MathlibAnnex.CStarAlgebra.CAR.shellFamilyLinks`

Supporting route explanation

Lean source signature (exact)

```
abbrev AtomicCounterexampleAlgebra := ShellFamilyTarget homogeneityShellFamily
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In the source	Mathematical meaning
AtomicCounterexampleAlgebra	The abbreviation denotes the one particular concrete algebra T in the statement, rather than an arbitrary algebra chosen anew for each representation.
homogeneityShellFamily	The fixed CAR shell family of automorphisms and partial isometries supplied by the proved homogeneity construction, based at the root state ω . It carries no additional KOS or CH hypothesis.
ShellFamilyTarget homogeneityShellFamily	The full RHS is $T = C^*(\pi(A) \cup \{L_i : i \in I\}) \subseteq B(\mathcal{H})$, with $\mathcal{H} = \bigoplus_{i \in I} H_i$, $\pi = \bigoplus_i \pi_i$, and the links chosen once for that family. The index set I need not be countable. Simplicity and capture are separately proved properties, not clauses of this abbreviation.

Further source notes: The separately cited `ShellFamilyTarget` and `shellFamilyLinks` explain its generated-algebra meaning; their definitions are not fragments of this abbreviation.

Earlier published Card and PDF

▼ Exact content identity

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